Types of Data in Computer Vision

1. Image data – Digital images (camera, satellite, medical scans, surveillance footage).
2. Video data – Video captured using camera.

Statistical Data

- Definition: Involves collecting, analyzing, interpreting data to discover insights.
- Types:
    - Numeric Data: Tables, Excel sheets.
    - Time Series Data: Data at specific intervals (stock prices, weather, sensor readings).
    - Textual Data: Documents, PDFs.
    - Visual Data: Images, videos.

Data Acquisition / Acquiring Data

- Definition: Process of gathering raw facts, figures, statistics from sources for AI projects.
- Steps:
    1. Data Discovery – Searching for new datasets.
    2. Data Augmentation – Adding more data or modifying existing data (flipping, rotation, brightness, occlusion).
    3. Data Generation – Generating synthetic data using sensors or simulations.

Sources of Data Acquisition

Primary Sources

- Collected directly via: Surveys, experiments, interviews, observations.

- Example: Names, addresses, phone numbers, percentages in an Excel sheet.

Secondary Sources

- Data collected from external/public sources.
- Examples:
    - **Kaggle** – Online data science community.
    - **Google Dataset Search** – Toolbox for finding datasets.
    - **UCI Repository** – Machine learning datasets.
    - Government portals, publications, market research.

Best Practices for Acquiring Data

1. Set Clear Goals – Define purpose, type, and detail level of data.
2. Identify Sources – Choose reliable primary/secondary sources.
3. Evaluate Sources – Ensure accuracy, relevance, and permissions.
4. Collect & Prepare – Fix errors, remove duplicates, clean data.
5. Validate & Store – Cross-check, update, and follow laws/regulations.

Good Data vs Bad Data

- Good Data: Structured, accurate, consistent, relevant, well-presented.
- Bad Data: Scattered, inaccurate, inconsistent, duplicate/missing values, irrelevant.

Data Acquisition from Websites

- Data Scraping: Collecting data using software (like web scraping).
- Legal if done with permission.

- Without permission = misuse (like picking fruit from someone else’s garden).

Ethical Concerns in Data Acquisition

1. Bias – Avoid preferences/partiality in data.
2. Consent – Take permission before collecting/using data.
3. Transparency – Clearly state how data will be used.
4. Anonymity – Protect identity of individuals.
5. Accountability – Take responsibility if data is misused.

1. Data Usability & Quality

- **Usability**: Data must be structured (tables, charts) to be meaningful.
- **Depends on:**
    - **Structure** → Organized format (Spreadsheet > Plain text).
    - **Cleanliness** → No duplicates, missing values, or outliers.
    - **Accuracy** → Matches real-world values (e.g., correct student marks).

Good Data = Structured + Clean + Accurate → Reliable analysis.
Bad Data = Unorganized + Dirty + Inaccurate → Misleading decisions.

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2. Features of Data

- **Feature** = Characteristic / Property in a dataset.
    - Example (Student data): Name, Age, Marks.
    - Example (Image data): Color, Brightness.

Types of Features

1. **Independent Features (Predictors)**
    - Input variables → used to make predictions.
    - Example: Hours studied, Attendance.
2. **Dependent Feature (Target)**
    - Output variable → predicted using inputs.
    - Example: Exam marks.

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3. Accuracy of Data

- **Accurate Data** → Consistent, correct values → correct conclusions.
    - Ex: Egg weights recorded correctly.
- **Inaccurate Data** → Errors → wrong conclusions.
    - Ex: Egg weights inconsistent due to mistakes.

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4. In AI Models

- **Independent Features (Cause/Input):** Study hours, Attendance, Previous grades.
- **Dependent Feature (Effect/Output):** Final exam grade.

Data Preprocessing

Preparing raw data for ML.
Steps:
1. Cleaning: Remove duplicates, fix errors/missing values.
2. Transformation: Convert to usable format.
3. Reduction: Reduce size, keep key info.
4. Integration/Normalization: Merge data, scale uniformly.
5. Feature Selection: Choose the most important variables.

Data Processing vs Interpretation(imp)

- Processing: Cleaning, training, predicting.
- Interpretation: Analyzing results to find patterns, refine models, and make decision

1. Data Processing

- Definition: Refining raw data for analysis.
- Purpose: Ensure data is accurate, relevant, and usable.
- Key Steps: Cleaning, organizing, summarizing.
- Importance: Helps computers understand data; critical for business, science, and tech.

2. Data Interpretation

- Definition: Making sense of processed data to find patterns and insights.
- Purpose: Inform decisions and answer key questions.

3. Interpretation Process (5 Steps)

1. Acquire: Gather raw data (surveys, sensors).
2. Process: Clean and organize data.
3. Analyze: Identify patterns (using stats/visualization).
4. Interpret: Derive actionable conclusions.
5. Present: Communicate findings (graphs, summaries).

Goal: Enable data-driven decisions.

Methods of Data Interpretation

1. Quantitative Interpretation

- Definition: Analyzing numeric data.
- Sources: Surveys, experiments.
- Purpose: Find patterns, trends.
- Techniques: Mean, median, standard deviation.
- Answers: "How many?", "How often?".

2. Qualitative Interpretation (Brief)

- Analyzes non-numeric data (e.g., text, opinions).

Data Collection Methods (Quantitative)

- Interviews (structured for numbers).

- Polls (single-question surveys).
- Observations (numeric tracking).
- Longitudinal Studies (track changes over time).
- Surveys (large-scale quantifiable data).

Qualitative Data Analysis

Steps:

1. Code Data: Assign labels (e.g., "ease of use") to identify patterns.
2. Analyze for Insights: Group codes into themes (e.g., "User Experience").
3. Report Insights: Present findings with quotes and visuals.

Qualitative vs Quantitative Interpretation

Aspect	Qualitative	Quantitative
Data Type	Categorical (descriptive)	Numerical (numbers & statistics)
Provides	Insights into feelings, reasons, context	Insights into quantity, frequency
Answers	"How?" and "Why?"	"When?", "How many?", "How often?"
Methods	Interviews, focus groups, observations	Surveys, polls, tests, assessments
Example Question	Why do students like online classes?	How many students like online classes?

Types of Data Interpretation

1. Textual

- Format: Words/paragraphs.
- Best for: Small, simple data.
- Limitation: Hard to follow with large data.
- Example: A paragraph summarizing sales increases.

2. Tabular

- Format: Rows and columns in a table.
- Advantage: Structured; easy to compare values.

- Key Feature: Includes a descriptive title.
- Example: A table of student marks with percentages.

3. **Graphical**

- Format: Visual charts/graphs.
- Advantage: Shows trends, comparisons, relationships easily.
- Common Types:
    - Bar Graph: Compare categories.
    - Line Graph: Show trends over time.
    - Pie Chart: Display parts of a whole.
    - Scatter Plot: Reveal relationships between variables.

Creating a Bar Graph in Tableau

1. Setup & Connect Data

- Prepare data in Excel (e.g., students.xlsx with columns: Student, Subject, Score).
- Open Tableau → Connect → Microsoft Excel → Select file → Open.
- Go to Sheet 1 to start.

2. Build Basic Bar Graph

- Drag Student to Columns.
- Drag Score to Rows.
- Drag Subject to Columns to create a clustered bar chart.
- Use sort icons (ascending/descending) to order bars.

3. Customize

- Color: Drag Subject to Color on Marks card.
- Labels: Click Label on Marks card → Check Show mark labels → Customize font/size.
- Chart Type: Use Show Me to switch chart types (e.g., packed bubbles).

4. Save & Share

- Duplicate Sheet: Right-click sheet tab → Duplicate.
- Save: File → Save As (or Export Packaged Workbook).
- Share Online: File → Save to Tableau Public/Server (requires account).